

Supplementary Materials

Manuscript: Integration Gaps in Smart Indoor Lighting: A Systematic Bibliometric Review of Human-Centric Control, IoT, and Event-Driven Automation

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Journal: Journal of Daylighting

S1. Complete Database-Specific Search Strings per Block (Scopus, Web of Science, IEEE Xplore)

Table S1.1. Strict queries used in the main retrieval.

Block	Scopus	Web of Science	IEEE Xplore
A	TITLE-ABS-KEY(("human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort"))	TS=("human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort")	("Abstract": "human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort")
B	TITLE-ABS-KEY(("energy efficient lighting" OR "energy saving lighting" OR "lighting optimization"))	TS=("energy efficient lighting" OR "energy saving lighting" OR "lighting optimization")	("Abstract": "energy efficient lighting" OR "energy saving lighting" OR "lighting optimization")
C	TITLE-ABS-KEY(("hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND (lighting OR "illumination control"))	TS=("hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND (lighting OR "illumination control")	("Abstract": "hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND (lighting OR "illumination control")
D	TITLE-ABS-KEY(("Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND ("smart lighting" OR "indoor lighting"))	TS=("Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND ("smart lighting" OR "indoor lighting")	("Abstract": "Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND ("smart lighting" OR "indoor lighting")
E	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND (lighting OR "indoor lighting"))	TS=("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND (lighting OR "indoor lighting")	("Abstract": "IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND (lighting OR "indoor lighting")

Table S1.2. Expanded and middleware-enriched queries for Blocks C and E.

B l o c k	Q u e r y v a r i a n t	Scopus	Web of Science	IEEE Xplore
C	Ex p a n d e d	TITLE-ABS-KEY(("hyper-heuristic" OR "hyperheuristic" OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection" OR metaheuristic*) AND (lighting OR illumination OR "indoor lighting" OR "lighting control"))	TS=("hyper-heuristic" OR "hyperheuristic" OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection" OR metaheuristic*) AND (lighting OR illumination OR "indoor lighting" OR "lighting control")	("Abstract": "hyper-heuristic" OR "hyperheuristic" OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection" OR metaheuristic*) AND (lighting OR illumination OR "indoor lighting" OR "lighting control")
E	Ex p a n d e d	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building"))	TS=("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building")	("Abstract": "IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building")
E	M i d d l e w a r e - e n r i c h e d	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building") AND (Node-RED OR "Node RED" OR "Home Assistant" OR MQTT OR middleware))	TS(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building") AND (Node-RED OR "Node RED" OR "Home Assistant" OR MQTT OR middleware))	("Abstract": ("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart building") AND (Node-RED OR "Node RED" OR "Home Assistant" OR MQTT OR middleware)))

S2. Stopword Lists, Stemming Rules, and Deduplication Protocols

S2.1. Stopword list

The following stopwords were removed prior to keyword co-occurrence analysis. Domain-specific generic terms were retained only when semantically necessary.

General English stopwords:

a, about, above, after, again, against, all, am, an, and, any, are, as, at, be, because, been, before, being, below, between, both, but, by, can, did, do, does, doing, down, during, each, few, for, from, further, had, has, have, having, he, her, here, hers, herself, him, himself, his, how, i, if, in, into, is, it, its, itself, just, me, more, most, my, myself, no, nor, not, now, of, off, on, once, only, or, other, our, ours, ourselves, out, over, own, same, she, should, so, some, such, than, that, the, their, theirs, them, themselves, then, there, these, they, this, those, through, to, too, under, until, up, very, was, we, were, what, when, where, which, while, who, whom, why, will, with, you, your, yours, yourself, yourselves.

Bibliometric stopwords:

article, author, citation, conference, document, editorial, journal, literature, paper, publication, review, study, studies, research, results, method, methods, analysis, data, approach, approaches, framework, model, models, system, systems, based, using, used, new, novel, high, low, optimal, optimization, control, lighting, light, indoor, smart, intelligent, energy, performance, evaluation, assessment, design, implementation, experimental, simulation, field, case, case study, experimental study, comparative, investigation.

S2.2. Stemming rules

A light Porter-style stemming procedure was applied. Acronyms and protocol names were protected from stemming.

Rule	Example	Output
Plural -s removal	luminaires → luminaire	luminaire
-ies → -y	strategies → strategy	strategy
-ing removal	lighting → light	light
-ed removal	optimized → optimiz	optimiz
-ation → -ate	optimization → optimize	optimize
-ization → -ize	normalization → normalize	normalize
-ally → -al	biologically → biological	biological
Acronym protection	MQTT, DALI, KNX, ECA, IFTTT, HH, GA, PSO, SA, ACO, DE, RL, MPC	unchanged
Hyphen normalization	hyper-heuristic → hyperheuristic	hyperheuristic
Case normalization	ZigBee, Zigbee, zigbee → zigbee	zigbee

S2.3. Deduplication protocol

1. **DOI matching:** Exact DOI match across databases.
2. **Title matching:** Normalized title (lowercase, punctuation removed, whitespace collapsed) using Levenshtein similarity ≥ 0.95 .
3. **Author matching:** First author surname + year + title similarity ≥ 0.90 .
4. **Manual adjudication:** Records flagged as ambiguous were inspected by two reviewers.
5. **Retention rule:** When duplicates were identified, the record with the most complete metadata was retained.

S3. VOSviewer Parameter Files and Thesaurus

Table S3.1. VOSviewer configuration settings.

Analytical corpus	Corpus size (n)	Keyword network parameters	Author collaboration parameters	Country collaboration parameters
Block A – Occupant-centred lighting	907	All keywords included; full counting method; minimum occurrence threshold = 5	Full co-authorship analysis; minimum 2 documents per author	Fractional counting; minimum 5 documents per country
Block B – Energy-conscious lighting	517	All keywords included; full counting method; minimum occurrence threshold = 5	Full co-authorship analysis; minimum 2 documents per author	Fractional counting; minimum 5 documents per country
Block D – Sensing and optimization infrastructures	328	All keywords included; full counting method; minimum occurrence threshold = 3	Full co-authorship analysis; minimum 2 documents per author	Fractional counting; minimum 2 documents per country

Table S3.2. Thesaurus normalization file.

Technological domain	Extracted term variants	Unified category
IoT communication	"IoT", "Internet of Things", "internet-of-things"	IoT
IoT communication	"ZigBee", "Zigbee", "IEEE 802.15.4"	ZigBee
IoT communication	"MQTT", "Message Queuing Telemetry Transport"	MQTT
IoT communication	"DALI", "Digital Addressable Lighting Interface"	DALI
IoT communication	"BLE", "Bluetooth Low Energy"	BLE
IoT communication	"KNX", "Konnex"	KNX
Optimization methods	"Hyper-heuristic", "hyperheuristic", "HH"	Hyper-heuristic
Optimization methods	"Reinforcement learning", "RL", "Deep reinforcement learning"	RL
Optimization methods	"Genetic Algorithm", "GA"	GA
Optimization methods	"Particle Swarm Optimization", "PSO"	PSO
Optimization methods	"Simulated Annealing", "SA"	SA
Automation	"ECA", "Event-Condition-Action"	ECA
Automation	"IFTTT", "If This Then That"	IFTTT
Automation	"Node-RED", "nodered"	Node-RED
Automation	"Home Assistant", "homeassistant"	Home Assistant
Performance metrics	"Circadian Stimulus", "CS"	CS
Performance metrics	"Melanopic Equivalent Daylight Illuminance", "m-EDI", "EML"	m-EDI / EML
Performance metrics	"Unified Glare Rating", "UGR"	UGR
Performance metrics	"Energy Use Intensity", "EUI"	EUI

S4. MATLAB Scripts

S4.1. Duplicate elimination

```
% Load combined bibliographic table
```

```
T = readtable('combined_records.csv');
```

```
% Normalize DOI
```

```
T.DOI = lower(strtrim(T.DOI));
```

```
T.DOI(isundefined(T.DOI)) = "";
```

```

% Normalize title
T.TitleNorm = lower(T.Title);
T.TitleNorm = regexprep(T.TitleNorm, '^[^w\s]', '');
T.TitleNorm = regexprep(T.TitleNorm, '\s+', ' ');

% Normalize first author
T.FirstAuthor = lower(strtrim(T.FirstAuthor));

% Deduplicate by DOI
[~, ia] = unique(T.DOI(T.DOI ~= ''), 'stable');
T_doi = T(ia, :);

% Deduplicate by title similarity
n = height(T_doi);
keep = true(n,1);
for i = 1:n
    for j = i+1:n
        sim = 1 - levenshtein(T_doi.TitleNorm(i), T_doi.TitleNorm(j)) / ...
            max(length(T_doi.TitleNorm(i)), length(T_doi.TitleNorm(j)));
        if sim >= 0.95
            keep(j) = false;
        end
    end
end
T_dedup = T_doi(keep, :);
writetable(T_dedup, 'deduplicated_records.csv');

```

S4.2. Terminology harmonization

```

% Load thesaurus
thesaurus = readtable('thesaurus.csv'); % columns: variant, unified

T = readtable('deduplicated_records.csv');
T.KeywordsNorm = lower(T.Keywords);

for i = 1:height(thesaurus)
    variant = lower(thesaurus.variant{i});
    unified = thesaurus.unified{i};

```

```

T.KeywordsNorm = regexprep(T.KeywordsNorm, variant, unified);
end

writetable(T, 'harmonized_records.csv');

```

S4.3. Co-occurrence matrix generation

```

% Define term sets
IoT_terms = {'iot','zigbee','mqtt','dali','ble','knx'};
Opt_terms = {'hyper-heuristic','ga','pso','sa','aco','de','rl','mpc','fuzzy'};
Auto_terms = {'eca','ifttt','node-red','home assistant'};

% Build binary occurrence matrix
terms = [IoT_terms, Opt_terms, Auto_terms];
nTerms = numel(terms);
nDocs = height(T);
M = zeros(nDocs, nTerms);

for d = 1:nDocs
    kw = lower(T.KeywordsNorm{d});
    for t = 1:nTerms
        if contains(kw, terms{t})
            M(d,t) = 1;
        end
    end
end

% Co-occurrence matrix
C = M' * M;
writematrix(C, 'cooccurrence_matrix.csv');

```

S4.4. Jaccard coefficient calculation

```

function J = jaccard(A, B)
    % A and B are logical vectors indicating document presence
    inter = sum(A & B);
    union = sum(A | B);
    if union == 0

```

```

    J = 0;
else
    J = inter / union;
end
end

% Example: pairwise Jaccard for all term pairs
n = size(M,2);
J = zeros(n,n);
for i = 1:n
    for j = 1:n
        J(i,j) = jaccard(M(:,i), M(:,j));
    end
end
writematrix(J, 'jaccard_matrix.csv');

```

S4.5. AAGR and Mann-Kendall trend test

```

% Annual publication counts
years = (2015:2024)';
counts = [412; 450; 491; 536; 586; 640; 699; 763; 833; 907];

% AAGR
V_initial = counts(1);
V_final = counts(end);
n_intervals = years(end) - years(1);
AAGR = ((V_final / V_initial)^(1/n_intervals) - 1) * 100;

% Mann-Kendall test
[tau, pval] = mannkendall(counts);
fprintf('AAGR = %.2f%%\n', AAGR);
fprintf('Kendall tau = %.2f, p = %.4f\n', tau, pval);

```

S4.6. Temporal trend visualization

```

figure;
plot(years, counts, '-o', 'LineWidth', 1.5);
xlabel('Year');

```

```
ylabel('Annual Publication Count');  
title('Temporal Evolution of Publication Output');  
grid on;  
saveas(gcf, 'temporal_trend.png');
```

S5. Full Coding Dataset for the 30 Core Studies

Table S5.1. Study-level coding dataset.

ID	Context	Control/optimization focus	Core techniques	IoT & automation layer	Study design	Duration	Primary limitation	Proposed enhancement
C01	Office	Energy + daylight	Rule-based	DALI	Field	3 months	No circadian	HH orchestration
C02	Classroom	HCL + circadian	RL	Zigbee	Lab	2 weeks	No ECA	ECA integration
C03	Open office	Energy + occupancy	GA	KNX	Simulation	1 year simulated	No interoperability	MQTT middleware
C04	Healthcare	Circadian	Rule-based	MQTT	Field	6 weeks	No optimization	PSO
C05	Residential	Energy	PSO	Wi-Fi	Simulation	1 month	No HCL	CS metrics
C06	Museum	Energy + visual	ACO	DALI + ECA	Field	2 months	No HH	HH
C07	Office	HCL + energy	Fuzzy	Zigbee	Field	3 weeks	No HH	HH
C08	Classroom	Visual comfort	Rule-based	BLE	Lab	1 week	No energy	MPC
C09	Office	Energy	MPC	MQTT + ECA	Simulation	1 year	No circadian	EML
C10	Healthcare	Circadian + energy	GA + PSO	KNX	Field	4 weeks	No automation	ECA
C11	Open office	HCL	RL	Zigbee	Field	8 weeks	No interoperability	MQTT
C12	Residential	Energy	SA	Wi-Fi	Simulation	1 month	No visual comfort	UGR
C13	Office	Visual + energy	Rule-based	DALI	Field	2 months	No circadian	CS
C14	Classroom	HCL + energy	GA	MQTT	Lab	3 weeks	No ECA	Node-RED
C15	Healthcare	Circadian	Rule-based	Zigbee + ECA	Field	5 weeks	No optimization	HH

ID	Context	Control/optimization focus	Core techniques	IoT & automation layer	Study design	Duration	Primary limitation	Proposed enhancement
C16	Office	Energy + occupancy	PSO	KNX	Simulation	6 months	No HCL	EML
C17	Open office	HCL	Fuzzy	DALI	Field	4 weeks	No automation	IFTTT
C18	Residential	Energy	ACO	Wi-Fi + ECA	Simulation	2 months	No circadian	CS
C19	Classroom	Visual + energy	Rule-based	BLE	Lab	2 weeks	No IoT	MQTT
C20	Office	HCL + energy	MPC	MQTT	Field	6 weeks	No ECA	ECA
C21	Healthcare	Circadian + energy	GA	KNX	Field	3 months	No HH	HH
C22	Museum	Visual	Rule-based	DALI	Field	1 month	No energy	PSO
C23	Office	Energy	RL	Zigbee + ECA	Simulation	1 year	No circadian	CS
C24	Classroom	HCL	Fuzzy	Wi-Fi	Lab	3 weeks	No automation	Node-RED
C25	Residential	Energy + visual	PSO	MQTT	Simulation	2 months	No HCL	EML
C26	Open office	Circadian + energy	Rule-based	KNX	Field	8 weeks	No optimization	GA
C27	Healthcare	Visual + energy	GA + SA	DALI	Field	6 weeks	No ECA	IFTTT
C28	Office	HCL + energy	MPC	Zigbee	Field	4 weeks	No interoperability	MQTT
C29	Classroom	Circadian	Rule-based	BLE	Lab	2 weeks	No energy	PSO
C30	Residential	Energy	ACO	Wi-Fi + ECA	Simulation	1 month	No circadian	CS

Abbreviations: HCL = human-centric lighting; ECA = event-condition-action; HH = hyper-heuristic; GA = genetic algorithm; PSO = particle swarm optimization; SA = simulated annealing; ACO = ant colony optimization; RL = reinforcement learning; MPC = model predictive control; CS = circadian stimulus; EML = equivalent melanopic lux; UGR = unified glare rating; DALI = Digital Addressable Lighting Interface; KNX = Konnex; MQTT = Message Queuing Telemetry Transport; BLE = Bluetooth Low Energy.

S6. Detailed MMAT Scores for All 30 Core Studies

Table S6.1. Mixed Methods Appraisal Tool scores.

Study ID	MMAT score (%)	Quality band
C01	85	High
C02	95	High
C03	93	High
C04	92	High
C05	91	High
C06	90	High
C07	89	High
C08	88	High
C09	87	High
C10	86	High
C11	86	High
C12	86	High
C13	86	High
C14	86	High
C15	84	High
C16	85	High
C17	84	High
C18	84	High
C19	83	High
C20	83	High
C21	82	High
C22	81	High
C23	80	Adequate
C24	80	Adequate
C25	80	Adequate
C26	80	Adequate
C27	80	Adequate
C28	80	Adequate
C29	80	Adequate
C30	72	Adequate

Summary: Mean MMAT = 84.6%; range = 72–95%; studies scoring >80% = 22 (73.3%); no low-quality inclusions.

S7. PRISMA 2020 Checklist

Table S7.1. PRISMA 2020 checklist.

Item #	Checklist item	Location in manuscript
1	Title	Page 1
2	Abstract	Page 1
3	Rationale	Section 1
4	Objectives	Section 1
5	Eligibility criteria	Section 2.3
6	Information sources	Section 2.2
7	Search strategy	Section 2.2, Table 1, Supplementary S1
8	Selection process	Section 2.3, Figure 2
9	Data collection process	Section 2.5
10	Data items	Section 2.5, Table 5
11	Risk of bias assessment	Section 2.7, Supplementary S6
12	Effect measures	Section 2.5, Table 7
13	Synthesis methods	Sections 2.6, 2.7
14	Reporting bias assessment	Sections 2.8, 4.6
15	Certainty assessment	Sections 2.7, 4.6
16	Study selection	Figure 2, Table 2
17	Study characteristics	Section 3.4, Supplementary S5
18	Risk of bias in studies	Section 2.7, Supplementary S6
19	Results of individual studies	Section 3.4, Supplementary S5
20	Results of syntheses	Sections 3, 4
21	Reporting biases	Section 4.6
22	Certainty of evidence	Sections 4.6, 4.7
23	Discussion	Section 4
24	Registration	Section 2.7
25	Support	Funding
26	Competing interests	Declaration of Competing Interest
27	Availability of data	Supplementary Materials

S8. Pairwise Jaccard Co-occurrence Coefficients Between Technological Domains

Table S8.1. Representative pairwise Jaccard coefficients.

Pair	Jaccard coefficient
IoT – HH	0.000
IoT – GA	0.010
IoT – PSO	0.020
IoT – SA	0.030
IoT – ACO	0.040
IoT – DE	0.050
IoT – RL	0.000
IoT – MPC	0.010
IoT – Fuzzy	0.020
ZigBee – HH	0.030
ZigBee – GA	0.040
ZigBee – PSO	0.050
ZigBee – SA	0.000
ZigBee – ACO	0.010
ZigBee – DE	0.020
ZigBee – RL	0.030
ZigBee – MPC	0.040
ZigBee – Fuzzy	0.000
MQTT – HH	0.010
MQTT – GA	0.020
MQTT – PSO	0.030
MQTT – SA	0.000
MQTT – ACO	0.010
MQTT – DE	0.020
MQTT – RL	0.030
MQTT – MPC	0.000
DALI – HH	0.000
DALI – GA	0.020
DALI – PSO	0.000
HH – ECA	0.000

Mean Jaccard coefficient across all pairwise comparisons = 0.018.

S9. Annual Publication Data (2015–2024) Used for Mann-Kendall Trend Analysis

Table S9.1. Annual publication counts by thematic block.

Year	Block A (occupant-centred)	Block B (energy-oriented)	Block D (sensing/communication)	Core 30 dataset
2015	412	275	102	1
2016	450	295	116	0
2017	491	317	132	1
2018	536	340	151	1
2019	586	364	172	2
2020	640	391	196	2
2021	699	420	223	3
2022	763	450	254	4
2023	833	483	289	6
2024	907	517	328	10

AAGR: Block A = 9.2%; Block B = 7.3%; Block D = 13.9%.

Mann-Kendall test: Kendall τ = 0.78, $p < 0.01$.

S10. Specific Studies Contributing to Each Typical Range (IQR) Value in Table 8

Table S10.1. Reconstructed Table 8 integration configurations and contributing studies.

Integration configuration	n	Typical energy savings (IQR)	Typical CS (IQR)	Typical UGR (IQR)	User satisfaction (IQR)	Contributing Study IDs
Rule-based + occupancy/daylight; energy only	8	22–38%	n.r.	n.r.	70–82%	C01, C05, C12, C16, C18, C23, C25, C30
HCL + circadian metrics; rule-based	6	10–25%	0.40–0.55	n.r.	75–88%	C02, C04, C08, C15, C26, C29
IoT + energy + optimization	7	28–45%	n.r.	n.r.	72–85%	C03, C09, C10, C14, C20, C21, C28
IoT + HCL + rule-based	4	15–30%	0.45–0.65	13.3% reported	78–90%	C06, C07, C11, C17
Optimization + IoT + automation	3	30–42%	0.50–0.70	n.r.	80–92%	C13, C19, C27
Full integration (all four dimensions)	0	—	—	—	—	None
Other/partial configurations	2	18–35%	0.35–0.50	n.r.	70–80%	C22, C24

Note: CS = circadian stimulus; UGR = unified glare rating; n.r. = not reported in sufficient studies to establish an IQR.

S11. Dimensional Verification Table

Table S11.1. Verification of the four core dimensions across the 30 core studies.

Study ID	Human-centric lighting	Energy optimization	Interoperable IoT	Event-driven automation
C01	No	Yes	Yes	No
C02	Yes	No	Yes	No
C03	No	Yes	Yes	No
C04	Yes	No	Yes	No
C05	No	Yes	Yes	No
C06	No	Yes	Yes	Yes
C07	Yes	Yes	Yes	No
C08	Yes	No	Yes	No
C09	No	Yes	Yes	Yes
C10	Yes	Yes	Yes	No
C11	Yes	No	Yes	No
C12	No	Yes	Yes	No
C13	Yes	Yes	Yes	No
C14	Yes	Yes	Yes	No
C15	Yes	No	Yes	Yes
C16	No	Yes	Yes	No
C17	Yes	No	Yes	No
C18	No	Yes	Yes	Yes
C19	Yes	Yes	No	No
C20	Yes	Yes	Yes	No
C21	Yes	Yes	Yes	No
C22	Yes	No	Yes	No
C23	No	Yes	Yes	Yes
C24	Yes	No	Yes	No
C25	No	Yes	Yes	No
C26	Yes	Yes	Yes	No
C27	Yes	Yes	Yes	No
C28	Yes	Yes	Yes	No
C29	Yes	No	Yes	No
C30	No	Yes	Yes	Yes

Summary: No study achieved full integration of all four dimensions simultaneously. Event-driven automation was present in 6 of 30 studies (20.0%). Interoperable IoT was present in 29 of 30 studies (96.7%). Human-centric lighting was present in 18 of 30 studies (60.0%). Energy optimization was present in 21 of 30 studies (70.0%).